

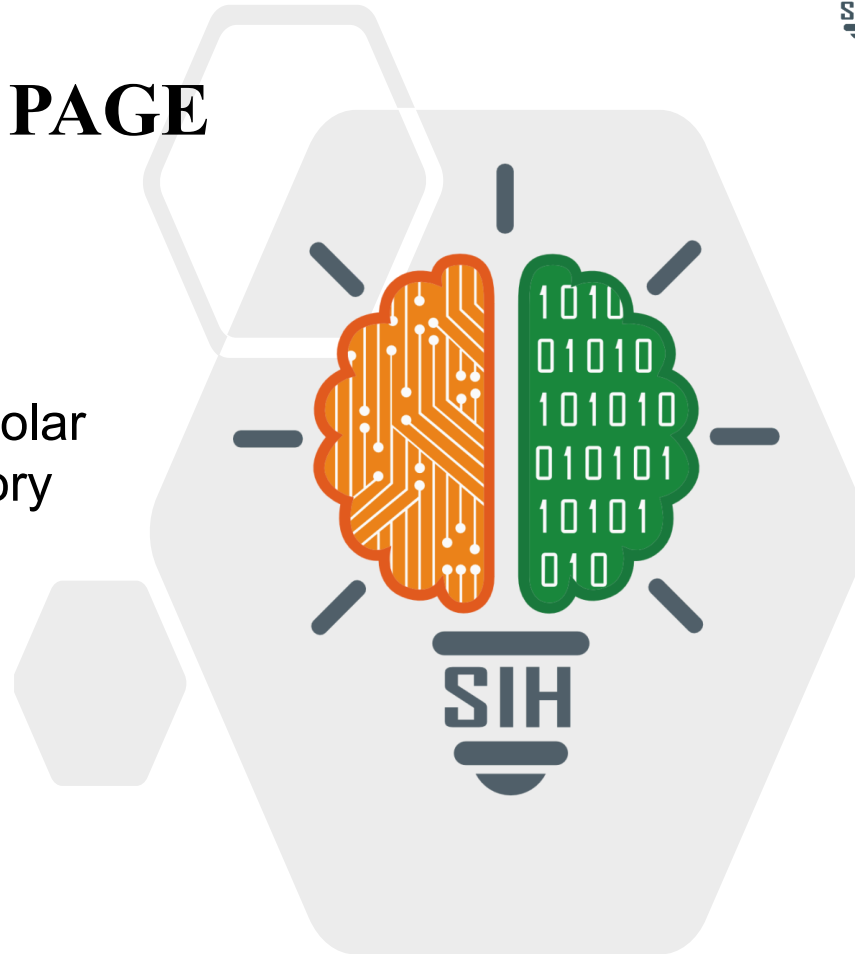
SMART INDIA HACKATHON 2026



SMART INDIA
HACKATHON
2026

TITLE PAGE

- **Problem Statement ID** – 26063
- **Problem Statement Title**- Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal
- **Theme**- Smart Education
- **PS Category**- Software
- **Team ID**- 182654
- **Team Name (Registered on portal)** - Claude Can Code



HIMVANI: Polar Knowledge Portal

THE PROBLEM

44 Antarctic expeditions of reports, photos and data sit in separate places, mostly untagged, shared in only two languages.

PROPOSED SOLUTION

One AI portal that archives every expedition, answers questions with cited sources, and turns polar science into approved outreach in 36 Indian languages.

01 Detailed explanation: how HIMVANI works

INPUT

Scientists & stations

Reports, photos, videos, datasets from Maitri, Bharati, Himadri

HIMVANI CORE

- 1 **Upload** by web, email or offline app
- 2 **Understand**: AI tags, OCR, transcripts
- 3 **Index** with source links, FAIR metadata

Ask

Any question in 36 languages, answered with citations

Studio

One update becomes kids, press and social drafts; a reviewer approves

Students & public
learn in their language

MoES outreach
posts on web & social

02 How it addresses the problem

- Split across 2 websites → **One search, one Ask box**
- Metadata forms skipped → **AI fills it on upload**
- Outreach in 2 languages → **Drafts in 36 languages**
- AI text can be wrong → **Every claim cited, human-approved**

03 Innovation and uniqueness

	NCPOR site	NPDC	AI chatbots	HIMVANI
Cited answers	X	X	X	✓
Auto-tagging	X	Manual	X	✓
Indian languages	2	1	Unverified	36
Approval workflow	Manual	X	X	✓

01 Methodology: system architecture

1 Sources

NCPOR reports Station photos & video NPDC datasets
News & press

2 Ingest

Web upload Email-to-archive Offline field app, syncs on link

3 AI enrichment

OCR Whisper transcripts CLIP image tags Auto metadata

4 Knowledge store

PostgreSQL + pgvector Object store FAIR metadata, DOIs

5 Services

Ask: RAG with citations Explore map Content Studio

6 Channels

Web portal X YouTube Instagram Bhashini: 36 languages

02 Technologies to be used

Interface Next.js, React PWA, CesiumJS, Leaflet

AI models RAG + LLM, Whisper, CLIP, OCR

Language Bhashini APIs, locked polar glossary

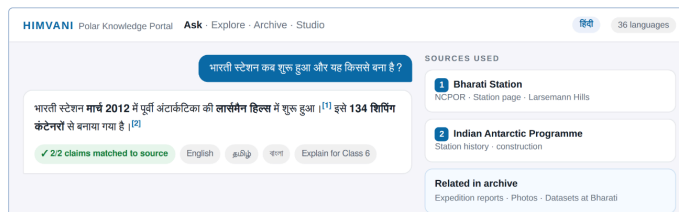
Data & API FastAPI, PostgreSQL + pgvector

Hosting NIC MeghRaj cloud

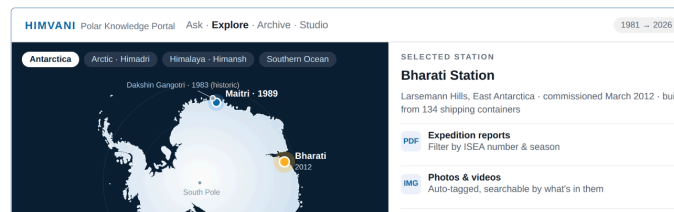
CHECKS AT EVERY STEP

Each answer sentence matched to a source · embargo dates
and access roles · a person approves before anything is
published

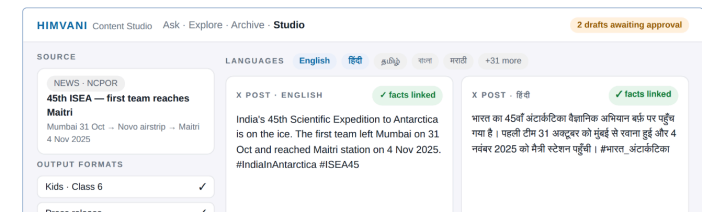
03 Working prototype screens from our MVP



Ask – Hindi question, answer with 2 sources



Explore – stations and expeditions on a map



Studio – one update, bilingual drafts to approve

01 Analysis of the feasibility of the idea



Technical

Open-source, proven parts:
pgvector, Whisper, CLIP



Data

NCPOR reports, photos and
NPDC datasets already public



Language

Bhashini is a government API
for 36 text languages



Team

Six of us: 2 frontend, 2 backend
+ AI, 1 data, 1 design

02 Viability and sustainability

₹0

licence cost, open-source stack

REACHOUT

fits PRITHVI's outreach sub-
scheme

MeghRaj

NIC cloud, run by NCPOR ICT

Reusable

by INCOIS, NCCR and IMD

03 12-week build plan week

1 2 3 4 5 6 7 8 9 10 11 12

Archive + Ask
upload, tagging, cited
answers

Studio + Bhashini
drafts, 36 languages,
approvals

Map + pilot
NPDC link, offline sync

Week 12: pilot with the NCPOR outreach team on one expedition update

04 Potential challenges and risks

Impact

AI states something wrong in public

High

Embargoed data goes public

High

Scientific terms mistranslated

Medium

Weak satellite links at stations

Medium

Scientists don't adopt another tool

Medium

05 Strategies for overcoming these challenges

Answers only from sources; claim check and human approval

Role-based access, embargo dates, public/internal flag per file

Locked polar glossary in all 36 languages, reviewer sign-off

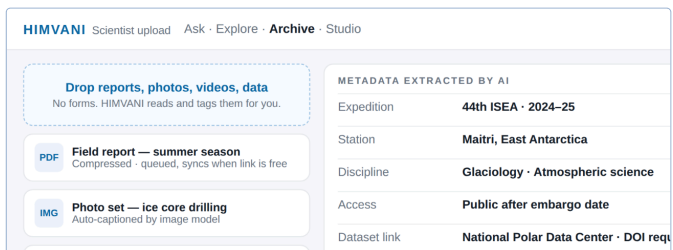
Offline-first app with compressed, resumable uploads

Zero-form upload and email-to-archive; AI fills metadata

IMPACT AND BENEFITS

01 Potential impact on the target audience

Scientists at NCPOR



HIMVANI Scientist upload Ask · Explore · Archive · Studio

Drop reports, photos, videos, data
No forms. HIMVANI reads and tags them for you.

PDF Field report — summer season
Compressed · queued, syncs when link is free

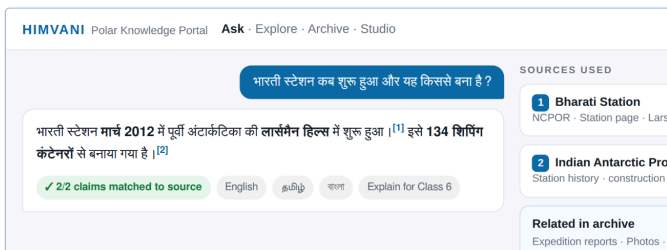
IMG Photo set — ice core drilling
Auto-captioned by image model

METADATA EXTRACTED BY AI

Expedition	44th ISEA · 2024–25
Station	Maitri, East Antarctica
Discipline	Glaciology · Atmospheric science
Access	Public after embargo date
Dataset link	National Polar Data Center · DOI req

Today files sit untagged, forms skipped
With us drop a file; it is findable, citable, linked to NPDC

Students and public



HIMVANI Polar Knowledge Portal Ask · Explore · Archive · Studio

भारती स्टेशन कब शुरू हुआ और यह किससे बना है?

भारती स्टेशन मार्च 2012 में पूर्वी अंटार्कटिका की लार्सन हिल्स में शुरू हुआ।^[1] इसे 134 शिपिंग कंटेनरों से बनाया गया है।^[2]

✓ 2/2 claims matched to source English தமிழ் বাংলা Explain for Class 6

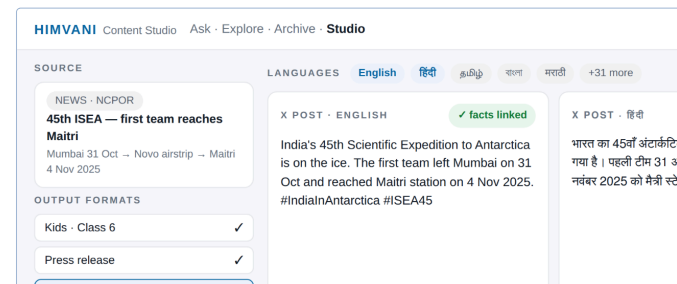
SOURCES USED

- 1 Bharati Station
NCPOR · Station page · Lars
- 2 Indian Antarctic Pro
Station history · construction

Related in archive
Expedition reports · Photos ·

Today English and Hindi pages, no way to ask
With us ask in Tamil or Hindi, get a sourced answer

Outreach cell, MoES



HIMVANI Content Studio Ask · Explore · Archive · Studio

SOURCE

NEWS · NCPOR

45th ISEA — first team reaches Maitri
Mumbai 31 Oct → Novo airstrip → Maitri
4 Nov 2025

OUTPUT FORMATS

Kids · Class 6 ✓

Press release ✓

LANGUAGES English हिंदी தமிழ் বাংলা मराठी +31 more

X POST · ENGLISH ✓ facts linked

India's 45th Scientific Expedition to Antarctica is on the ice. The first team left Mumbai on 31 Oct and reached Maitri station on 4 Nov 2025. #IndiaInAntarctica #SEA45

X POST · हिंदी

भारत का 45वाँ अंटार्कटिक गमन है। पहली टीम 31 अक्टूबर 2025 को मुंबई स्टेशन से मैत्री स्टेशन के लिए निकली।

Today each post hand-written, one language
With us one upload gives 216 drafts to review

REACH OF ONE EXPEDITION UPDATE

Languages

Today 2

HIMVANI 36

Drafts per update (6 formats × 36)

Today 1

HIMVANI 216

02 Benefits of the solution

Social · SDG 4

Polar science for learners beyond English and Hindi

Environmental · SDG 13

Climate literacy on how polar ice shapes India's weather

Economic

No lost or redone field data; zero licence cost

Scientific

FAIR datasets with DOIs, easier to reuse worldwide

RESEARCH AND REFERENCES

1981–82First Indian expedition reached
Antarctica [3]**45th**expedition underway, 44 completed
[1, 2]**36**Indian text languages on Bhashini
[8]**₹4,797 cr**

PRITHVI scheme, 2021–26 [7]

01 Details / Links of the reference and research work

- 1 NCPOR – 45th ISEA team reaches Maitri (Nov 2025)
ncaor.gov.in/news/view/929
- 2 NCPOR – Call for 46th ISEA proposals (44 completed)
isea.ncpor.res.in/forms/46-ISEA-Webpage-Advertisment.pdf
- 3 PIB – Antarctica Expeditions (1981 start, Maitri, Bharati)
pib.gov.in/newsite/PrintRelease.aspx?relid=123510
- 4 Indian Antarctic Programme (Bharati, 134 containers)
en.wikipedia.org/wiki/Indian_Antarctic_Programme
- 5 NCPOR – Himadri, India's Arctic station
ncpor.res.in/.../340-himadri-station
- 6 National Polar Data Center, NCPOR
npdc.ncpor.res.in
- 7 PIB – PRITHVI scheme, ₹4,797 cr (incl. REACHOUT)
pib.gov.in/PressReleaselframePage.aspx?PRID=2003592
- 8 PIB – BHASHINI: 36 text, 23 voice languages (2026)
pib.gov.in/PressReleasePage.aspx?PRID=2267632
- 9 Wilkinson et al., FAIR data principles (2016)
doi.org/10.1038/sdata.2016.18
- 10 Lewis et al., RAG (2020); Radford et al., Whisper (2022)
arxiv.org/abs/2005.11401 arxiv.org/abs/2212.04356